

## What this firmware adds

Built on kachurovskiy/nanoels H4 at commit **5de1419**. Everything the stock firmware does is still here and works the same way — the list below is what is new. Upstream is 3,074 lines; this adds 9,438 across 21 files, about half of it tests and documentation.

| Spindle encoder                     |                                                                                                                                                                                 | Readouts                          |                                                                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Four times the resolution</b>    | Stock reads one channel at two edges per pulse. This reads both channels at four, so a 1000 PPR encoder resolves 4000 positions instead of 2000 — 0.09° rather than 0.18°.      | <b>Large position display</b>     | Z and X in two-row digits, readable across the shop.                                                                                  |
| <b>Direction in software</b>        | Reverses the count from the menu instead of swapping the A and B wires.                                                                                                         | <b>Diameter mode for X</b>        | Type and read diameters rather than radii.                                                                                            |
| <b>Belt and gear ratios</b>         | Spindle-to-encoder teeth are a setting, so a geared encoder no longer has to be folded into the PPR figure by hand.                                                             | <b>Spindle indexing</b>           | Divides a revolution into marks and beeps as you reach each one.                                                                      |
| <b>Divider and dead-band shape</b>  | Two ways to quieten a fluttering encoder. The dead-band can filter both directions rather than only the reverse.                                                                | <b>Constant cutting speed</b>     | Says what rpm the current diameter wants, from a material and tool table. Advisory: there is no spindle output on this board.         |
| <b>Live signal quality</b>          | Coherence compares how far the counter travelled against how far it moved. A dithering encoder reads as a stopped spindle on every other readout; this is the one that sees it. | <b>Configuration</b>              |                                                                                                                                       |
| <b>New modes</b>                    |                                                                                                                                                                                 | <b>Settings menu</b>              | 90 items on the panel. Stock has none — you edited the sketch and reflashed.                                                          |
| <b>XGEAR — power cross-feed</b>     | The electronic gearbox driving the cross slide instead of the carriage, for powered facing.                                                                                     | <b>Per-mode settings</b>          | Passes, spring passes, clearance and the rest belong to the operation.                                                                |
| <b>SLOT — the lathe as a shaper</b> | The spindle is not involved. Z strokes back and forth while X steps down, for keyways and flats.                                                                                | <b>Calibration routines</b>       | Fourteen guided measurements — screw pitch, backlash, travel, max speed, encoder PPR and direction — that measure rather than assume. |
| <b>TPR — tapered threading</b>      | X drifts across as Z advances, so the thread grows on a cone. NPT and BSPT.                                                                                                     | <b>machine_config.h</b>           | Every pin and machine constant in one file, out of the sketch.                                                                        |
| <b>Cutting</b>                      |                                                                                                                                                                                 | <b>Serial interface</b>           | \$ dumps every setting for backup; \$key=value writes one.                                                                            |
| <b>Spring passes</b>                | Repeats the final pass at depth to take out deflection.                                                                                                                         | <b>Connectivity</b>               |                                                                                                                                       |
| <b>Flank infeed</b>                 | Phases the tool along the helix so it cuts mostly on the leading flank instead of plunging on both.                                                                             | <b>WiFi access point</b>          | The controller makes its own network. Off by default: no radio at all.                                                                |
| <b>Peck parting</b>                 | Retracts periodically to break the chip.                                                                                                                                        | <b>Web configuration</b>          | Every setting from a phone at the lathe, plus a live status strip.                                                                    |
| <b>Thread database</b>              | 52 presets — metric, UN, BSPP, trapezoidal, ACME and NPT.                                                                                                                       | <b>Firmware over the air</b>      | Update without a laptop. Refused unless the machine is stopped.                                                                       |
| <b>One-key X retract</b>            | Pulls the tool clear and puts it back on the same number.                                                                                                                       | <b>Inputs and safety</b>          |                                                                                                                                       |
|                                     |                                                                                                                                                                                 | <b>Joystick</b>                   | Optional stick on the auxiliary terminals for jogging.                                                                                |
|                                     |                                                                                                                                                                                 | <b>Terminal conflict checking</b> | Four devices want the same six pins. Enabling one over another's is refused rather than quietly configuring the pins twice.           |
|                                     |                                                                                                                                                                                 | <b>Runtime enable</b>             | Handwheels and the stick switch on without a restart.                                                                                 |
|                                     |                                                                                                                                                                                 | <b>Distinct buzzer patterns</b>   | A refused key, a finished pass and a lost thread sound different.                                                                     |
|                                     |                                                                                                                                                                                 | <b>Underneath</b>                 |                                                                                                                                       |
|                                     |                                                                                                                                                                                 | <b>Host test suite</b>            | 777 checks on the arithmetic and 42 on the web page, run on a PC without a board. The firmware calls the same code the tests do.      |

## Coming from the stock firmware

Nothing here changes how the stock features behave, and your stored settings survive the update. Five things are worth doing once.

- 1 Take a backup first. Send \$ over USB and keep the text — there is no undo in the settings menu.
- 2 Leave the encoder **PPR** as it is. It still means pulses per revolution as marked; the firmware reads four counts from each rather than two, and works the rest out itself.
- 3 Check the **glitch filter**. The counter now sees twice the traffic, and the usable ceiling is **2.4e9 / (PPR × filter)** encoder rpm — a geared-up encoder spins faster than the spindle, so account for the belt too.
- 4 Run **Encoder signal** from the calibration menu and read the coherence figure during a real cut. It is the one readout that can tell you the cable is the problem.
- 5 Everything that used to mean editing the sketch is now in the settings menu. **machine\_config.h** only sets the defaults for a controller that has never been configured.

## Everything it does

| Feeding                          |                                                                   | What you can see             |                                                                        |
|----------------------------------|-------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------|
| <b>Electronic lead screw</b>     | Carriage feeds from the spindle at any pitch, metric or imperial. | <b>Position readout</b>      | Z and X, metric or inch, radius or diameter.                           |
| <b>Power cross-feed</b>          | The same, driving X.                                              | <b>Large display</b>         | Two-row digits for reading across the shop.                            |
| <b>Async feed</b>                | A fixed feed rate with no spindle involved.                       | <b>Spindle angle and rpm</b> | Plus surface speed at the tool.                                        |
| <b>Cone</b>                      | Constant ratio between X and Z for continuous taper turning.      | <b>Constant speed target</b> | The rpm the current diameter wants for the material set.               |
| <b>Manual jogging</b>            | Arrow keys, with a selectable step down to 0.01mm.                | <b>Spindle indexing</b>      | Divide a revolution into marks, with a beep at each.                   |
| <b>Handwheels</b>                | Up to two pulse generators, each assignable to an axis.           | <b>Encoder signal health</b> | Coherence and its low-water mark, live.                                |
|                                  |                                                                   | <b>Pass progress</b>         | Which pass of how many, with a bar.                                    |
| Automated operations             |                                                                   | Setting it up                |                                                                        |
| <b>Turning and facing</b>        | Multi-pass to a diameter or a face, returning between passes.     | <b>Settings menu</b>         | Everything adjustable on the panel — no reflashing.                    |
| <b>Parting and grooving</b>      | Feeds X in, with optional pecking.                                | <b>Calibration routines</b>  | Fourteen guided measurements on the actual lathe.                      |
| <b>Threading</b>                 | Single-point, multi-pass, multi-start, straight or tapered.       | <b>Thread database</b>       | 52 presets across six thread families.                                 |
| <b>Slotting</b>                  | Shaper strokes for keyways and flats.                             | <b>Backup and restore</b>    | One text dump over USB or from the browser.                            |
| <b>Half-spheres and ellipses</b> | Convex or concave, over multiple passes.                          | <b>Web interface</b>         | The same settings from a phone, with live status.                      |
| <b>G-code</b>                    | Stored programs sent from the browser or over USB.                | <b>Firmware over the air</b> | No laptop at the machine.                                              |
| <b>Fourth axis</b>               | An optional A1 axis, rotary or linear.                            |                              |                                                                        |
| Precision                        |                                                                   | Looking after itself         |                                                                        |
| <b>Backlash compensation</b>     | Per axis, taken up automatically on a reversal.                   | <b>Emergency stop</b>        | On a lost position, a key stuck at power-up, or a demanded overtravel. |
| <b>Soft limits</b>               | Four stops that define the cut and cannot be passed.              | <b>Refusals with reasons</b> | A rejected input says why on screen and sounds its own tone.           |
| <b>Gear and pulley ratios</b>    | Belt drives on any axis and on the encoder.                       | <b>Terminal conflicts</b>    | Two devices cannot claim the same pins.                                |
| <b>Spring passes</b>             | Repeat at depth to take out deflection.                           | <b>Motion lock on writes</b> | A setting cannot change underneath a move in progress.                 |
| <b>Flank infeed</b>              | Cut on the leading flank rather than plunging.                    |                              |                                                                        |

## Choosing hardware for a conversion

General guidance rather than a parts list — the right answer depends on the lathe. Two things are worth more attention than they usually get: the encoder cable, and whether your steppers can be trusted not to lose steps.

| Part                   | What to look for                                                                                                                                                                                                                                                                                                                                 |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Controller</b>      | The NanoEls H4 board — an ESP32-S3 with two stepper outputs, an encoder input, six auxiliary terminals and a 20×4 display. Boards and the PCB files are on the upstream project.                                                                                                                                                                 |
| <b>Spindle encoder</b> | An incremental A/B quadrature encoder, 600–1000 PPR. Mount it on the spindle directly or on a 1:1 belt if you can. Push-pull output is easier than open-collector. Gearing it up multiplies the pulse rate, and the usable ceiling is <b>2.4e9 / (PPR × glitch filter)</b> encoder rpm — check that figure before choosing a high-count encoder. |
| <b>Encoder cable</b>   | Shielded, twisted pairs, shield grounded at one end only, run away from the VFD and motor leads. This matters more than the encoder does: the coherence readout exists because noise here is the usual cause of a bad thread.                                                                                                                    |
| <b>Stepper motors</b>  | Size to the axis, not the lathe. Z has to pull the carriage and wants the larger motor; X moves far less mass. Closed-loop steppers cost more and remove the one failure this firmware cannot detect — a step commanded and not taken.                                                                                                           |
| <b>Drivers</b>         | Microstepping of 400–800 steps per screw revolution is plenty; more costs torque and buys nothing the lead screw can deliver. Set <b>Hold when idle</b> to off for open-loop drivers that run hot holding position.                                                                                                                              |
| <b>Lead screws</b>     | Ball screws remove most backlash and are worth it on X. Trapezoidal screws work — backlash is compensated in software, but compensation cannot help a screw that is also worn unevenly. Measure it with the backlash routine.                                                                                                                    |
| <b>Power</b>           | 24–48V for the steppers, sized for both at once. The controller runs from 5V.                                                                                                                                                                                                                                                                    |

Run the calibration routines after any mechanical change — the firmware only knows what it has been told or has measured, and a screw or motor swap makes every stored figure a guess.